

In terms of (1.4.8) and (1.4.9), the variational equation takes the form

$$a(w, u) = (w, f) + w(0)/l \quad (1.4.10)$$

Here, $a(\cdot, \cdot)$ and $(\cdot, \cdot)$ are examples of *symmetric, bilinear forms*. What this means is as follows: Let c_1 and c_2 be constants and let u , v , and w be functions. Then the *symmetry* property is

$$a(u, v) = a(v, u) \quad (1.4.11)$$

$$(u, v) = (v, u) \quad (1.4.12)$$

Bilinearity means linearity in each "slot"; for example,

$$a(c_1u + c_2v, w) = c_1a(u, w) + c_2a(v, w) \quad (1.4.13)$$

$$(c_1u + c_2v, w) = c_1(u, w) + c_2(v, w) \quad (1.4.14)$$

Exercise 1. Use the definitions of $a(\cdot, \cdot)$ and $(\cdot, \cdot)$ to verify the properties of symmetry and bilinearity.

The above notations are very concise; at the same time they capture essential mathematical features and thus are conducive to a mathematical understanding of variational and finite element methods. Diverse classes of physical problems can be written in essentially similar fashion to (1.4.10). Thus ideas developed and results obtained are seen at once to have very broad applicability.

1.5 GALERKIN'S APPROXIMATION METHOD

We shall now describe a method of obtaining approximate solutions to boundary-value problems based upon weak formulations. Our introduction to this subject is somewhat of an abstract treatment. However, the meaning should be significantly reinforced by the remaining sections of the chapter. It may be worthwhile for the reader to consult this section again after completing the rest of the chapter to make sure a full comprehension of the material is attained.

The first step in developing the method is to construct finite-dimensional approximations of $\mathcal{S}$ and $\mathcal{U}$. These collections of functions are denoted by $\mathcal{S}^h$ and $\mathcal{U}^h$, respectively. The superscript refers to the association of $\mathcal{S}^h$ and $\mathcal{U}^h$ with a *mesh*, or *discretization*, of the domain Ω , which is parameterized by a characteristic length scale h . We wish to think of $\mathcal{S}^h$ and $\mathcal{U}^h$ as being subsets of $\mathcal{S}$ and $\mathcal{U}$, respectively. This is written as

$$\mathcal{S}^h \subset \mathcal{S} \quad (\text{i.e., if } u^h \in \mathcal{S}^h, \text{ then } u^h \in \mathcal{S}) \quad (1.5.1)$$

$$\mathcal{U}^h \subset \mathcal{U} \quad (\text{i.e., if } w^h \in \mathcal{U}^h, \text{ then } w^h \in \mathcal{U}) \quad (1.5.2)$$

where the precise meaning is given in parentheses.² Consequences of (1.5.1) and (1.5.2) are (respectively) that if $u^h \in \mathcal{S}^h$ and $w^h \in \mathcal{U}^h$, then

$$u^h(1) = q \quad (1.5.3)$$

$$w^h(1) = 0 \quad (1.5.4)$$

The collections, $\mathcal{S}$, $\mathcal{U}$, $\mathcal{S}^h$, and $\mathcal{U}^h$, are often referred to as *function spaces*. The terminology *space* in mathematics usually connotes a linear structure. This has the following meaning: If c_1 and c_2 are constants and v and w are in $\mathcal{U}$, then $c_1 v + c_2 w$ is also in $\mathcal{U}$. Both $\mathcal{U}$ and $\mathcal{U}^h$ are thus seen to possess the property of a linear space. However, this property is clearly not shared by $\mathcal{S}$ and $\mathcal{S}^h$ due to the inhomogeneous boundary condition. For example, if u_1 and u_2 are members of $\mathcal{S}$, then $u_1 + u_2 \notin \mathcal{S}$, since $u_1(1) + u_2(1) = q + q = 2q$ in violation of the definition of $\mathcal{S}$. Nevertheless, the terminology function space is still (loosely) applied to $\mathcal{S}$ and $\mathcal{S}^h$.

(Bubnov-) Galerkin Method

Assume the collection $\mathcal{U}^h$ is given. Then, to each member $v^h \in \mathcal{U}^h$, we construct a function $u^h \in \mathcal{S}^h$ by

$$u^h = v^h + q^h \quad (1.5.5)$$

where q^h is a *given* function satisfying the essential boundary condition, i.e.,

$$q^h(1) = q \quad (1.5.6)$$

Note that (1.5.5) satisfies the requisite boundary condition also:

$$\begin{aligned} u^h(1) &= v^h(1) + q^h(1) \\ &= 0 + q \end{aligned} \quad (1.5.7)$$

Thus (1.5.5) constitutes a definition of $\mathcal{S}^h$; that is, $\mathcal{S}^h$ is all functions of the form (1.5.5). The key point to observe is that, up to the function q^h , $\mathcal{S}^h$ and $\mathcal{U}^h$ are composed of *identical* collections of functions. This property will be shown later on to have significant consequences for certain classes of problems.

We now write a variational equation, of the form of (1.4.10), in terms of $w^h \in \mathcal{U}^h$ and $u^h \in \mathcal{S}^h$:

$$a(w^h, u^h) = (w^h, f) + w^h(0)h \quad (1.5.8)$$

This equation is to be thought of as defining an approximate (weak) solution, u^h .

²This condition may be considered standard. However, it is often violated in practice. Strang [2] coined the terminology “variational crimes” to apply to this, and other, situations in which the classical rules of variational methods are violated. Many “variational crimes” have been given a rigorous mathematical basis (e.g., see [2]). We shall have more to say about this subject in subsequent chapters.

Substitution of (1.5.5) into (1.5.8), and the bilinearity of $a(\cdot, \cdot)$ enables us to write

$$a(w^h, v^h) = (w^h, \ell) + w^h(0)h - a(w^h, q^h) \quad (1.5.9)$$

The right-hand side consists of the totality of terms associated with given data (i.e., ℓ , q , and h). Equation (1.5.9) is to be used to define v^h , the unknown part of u^h .

The (Bubnov-) Galerkin form of the problem, denoted by (G) , is stated as follows:

$$(G) \begin{cases} \text{Given } \ell, q, \text{ and } h, \text{ as before, find } u^h = v^h + q^h, \text{ where } v^h \in \mathcal{U}^h, \\ \text{such that for all } w^h \in \mathcal{U}^h \\ \quad a(w^h, v^h) = (w^h, \ell) + w^h(0)h - a(w^h, q^h) \end{cases}$$

Note that (G) is just a version of (W) posed in terms of a finite-dimensional collection of functions, namely, $\mathcal{U}^h$.

To make matters more specific, q^h and $\mathcal{U}^h$ have to be explicitly defined. Before doing this, it is worthwhile to mention a larger class of approximation methods, called **Petrov-Galerkin methods**, in which v^h is contained in a collection of functions other than $\mathcal{U}^h$. Recent attention has been paid to methods of this type, especially in the context of fluid mechanics. For the time being, we will be exclusively concerned with the Bubnov-Galerkin method. The Bubnov-Galerkin method is commonly referred to as simply the Galerkin method, terminology we shall adopt henceforth. Equation (1.5.9) is sometimes referred to as the **Galerkin equation**.

Approximation methods of the type considered are examples of so-called **weighted residual methods**. The standard reference dealing with this subject is Finlayson [3]. For a more succinct presentation containing an interesting historical account, see Finlayson and Scriven [4].

1.6 MATRIX EQUATIONS; STIFFNESS MATRIX K

The Galerkin method leads to a coupled system of linear algebraic equations. To see this we need to give further structure to the definition of $\mathcal{U}^h$. Let $\mathcal{U}^h$ consist of all linear combinations of given functions denoted by $N_A : \bar{\Omega} \rightarrow \mathbb{R}$, where $A = 1, 2, \dots, n$. By this we mean that if $w^h \in \mathcal{U}^h$, then there exist constants c_A , $A = 1, 2, \dots, n$, such that

$$\begin{aligned} w^h &= \sum_{A=1}^n c_A N_A \\ &= c_1 N_1 + c_2 N_2 + \dots + c_n N_n \end{aligned} \quad (1.6.1)$$

The N_A 's are referred to as **shape**, **basis**, or **interpolation** functions. We require that each N_A satisfies

$$N_A(1) = 0, \quad A = 1, 2, \dots, n \quad (1.6.2)$$

from which it follows by (1.6.1) that $w^h(1) = 0$, as is necessary. $\mathcal{U}^h$ is said to have dimension n , for obvious reasons.

To define members of $\mathcal{S}^h$ we need to specify q^h . To this end, we introduce another shape function, $N_{n+1} : \bar{\Omega} \rightarrow \mathbb{R}$, which has the property

$$N_{n+1}(1) = 1 \quad (1.6.3)$$

(Note $N_{n+1} \notin \mathcal{U}^h$.) Then q^h is given by

$$q^h = q N_{n+1} \quad (1.6.4)$$

and thus

$$q^h(1) = q \quad (1.6.5)$$

With these definitions, a typical $u^h \in \mathcal{S}^h$ may be written as

$$\begin{aligned} u^h &= v^h + q^h \\ &= \sum_{A=1}^n d_A N_A + q N_{n+1} \end{aligned} \quad (1.6.6)$$

where the d_A 's are constants and from which it is apparent that $u^h(1) = q$.

Substitution of (1.6.1) and (1.6.6) into the Galerkin equation yields

$$\begin{aligned} a\left(\sum_{A=1}^n c_A N_A, \sum_{B=1}^n d_B N_B\right) &= \left(\sum_{A=1}^n c_A N_A, \ell\right) + \left[\sum_{A=1}^n c_A N_A(0)\right] h \\ &\quad - a\left(\sum_{A=1}^n c_A N_A, q N_{n+1}\right) \end{aligned} \quad (1.6.7)$$

By using the bilinearity of $a(\cdot, \cdot)$ and $(\cdot, \cdot)$, (1.6.7) becomes

$$0 = \sum_{A=1}^n c_A G_A \quad (1.6.8)$$

where

$$G_A = \sum_{B=1}^n a(N_A, N_B) d_B - (N_A, \ell) - N_A(0) h + a(N_A, N_{n+1}) q \quad (1.6.9)$$

Now the Galerkin equation is to hold for all $w^h \in \mathcal{U}^h$. By (1.6.1), this means for all c_A 's, $A = 1, 2, \dots, n$. Since the c_A 's are arbitrary in (1.6.8), it necessarily follows that each G_A , $A = 1, 2, \dots, n$, must be identically zero, i.e., from (1.6.9)

$$\sum_{B=1}^n a(N_A, N_B) d_B = (N_A, \ell) + N_A(0) h - a(N_A, N_{n+1}) q$$

(1.6.10)

Note that everything is known in (1.6.10) except the d_B 's. Thus (1.6.10) constitutes a system of n equations in n unknowns. This can be written in a more concise form as follows:

Let

$$K_{AB} = a(N_A, N_B) \quad (1.6.11)$$

$$F_A = (N_A, f) + N_A(0)h - a(N_A, N_{n+1})q \quad (1.6.12)$$

Then (1.6.10) becomes

$$\sum_{B=1}^n K_{AB} d_B = F_A, \quad A = 1, 2, \dots, n \quad (1.6.13)$$

Further simplicity is gained by adopting a matrix notation. Let

$$K = [K_{AB}] = \begin{bmatrix} K_{11} & K_{12} & \cdots & K_{1n} \\ K_{21} & K_{22} & \cdots & K_{2n} \\ \vdots & \vdots & & \vdots \\ K_{n1} & K_{n2} & \cdots & K_{nn} \end{bmatrix} \quad (1.6.14)$$

$$F = \{F_A\} = \begin{Bmatrix} F_1 \\ F_2 \\ \vdots \\ F_n \end{Bmatrix} \quad (1.6.15)$$

and

$$d = \{d_B\} = \begin{Bmatrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{Bmatrix} \quad (1.6.16)$$

Now (1.6.13) may be written as

$$Kd = F \quad (1.6.17)$$

The following terminologies are frequently applied, especially when the problem under consideration pertains to a mechanical system:

K = stiffness matrix

F = force vector

d = displacement vector

A variety of physical interpretations are of course possible.

At this point, we may state the matrix equivalent, (M) , of the Galerkin problem.

$$(M) \left\{ \begin{array}{l} \text{Given the coefficient matrix } K \text{ and vector } F, \text{ find } d \text{ such that} \\ Kd = F \end{array} \right.$$

The solution of (M) is, of course, just $d = K^{-1}F$ (assuming the inverse of K , K^{-1} , exists). Once d is known, the solution of (G) may be obtained at any point $x \in \bar{\Omega}$ by employing (1.6.6), viz.,

$$u^h(x) = \sum_{A=1}^n d_A N_A(x) + q N_{n+1}(x) \quad (1.6.18)$$

Likewise, derivatives of u^h , if required, may be obtained by term-by-term differentiation. It should be emphasized, that the solution of (G) is an *approximate* solution of (W) . Consequently, the differential equation and natural boundary condition are only approximately satisfied. The quality of the approximation depends upon the specific choice of N_A 's and the number n .

Remarks

1. The matrix K is symmetric. This follows from the symmetry of $a(\cdot, \cdot)$ and use of Galerkin's method (i.e., the same shape functions are used for the variations and trial solutions):

$$\begin{aligned} K_{AB} &= a(N_A, N_B) \\ &= a(N_B, N_A) \\ &= K_{BA} \end{aligned} \quad (1.6.19)$$

In matrix notation

$$K = K^T \quad (1.6.20)$$

where the superscript T denotes transpose. The symmetry of K has important computational consequences.

2. Let us schematically retrace the steps leading to the matrix problem, as they are typical of the process one must go through in developing a finite element method for any given problem:

$(S) \Leftrightarrow (W) \approx (G) \Leftrightarrow (M)$

(1.6.21)

The only apparent approximation made thus far is in approximately solving (W) via (G) . In more complicated situations, encountered in practice, the number of approximations increases. For example, the data f , q , and h may be approximated, as well as the domain Ω , calculation of integrals, and so on. Convergence proofs and error analyses involve consideration of each approximation.

3. It is sometimes convenient to write

$$u^h(x) = \sum_{A=1}^{n+1} N_A(x) d_A \quad (1.6.22)$$

where $d_{n+1} = q$.

1.7 EXAMPLES: 1 AND 2 DEGREES OF FREEDOM

In this section we will carry out the detailed calculations involved in formulating and solving the Galerkin problem. The functions employed are extremely simple, thus expediting computations, but they are also primitive examples of typical finite element functions.

Example 1 (1 degree of freedom)

In this case $n = 1$. Thus $w^h = c_1 N_1$ and $u^h = v^h + q^h = d_1 N_1 + q N_2$. The only unknown is d_1 . The shape functions must satisfy $N_1(1) = 0$ and $N_2(1) = 1$ (see (1.6.2) and (1.6.3)). Let us take $N_1(x) = 1 - x$ and $N_2(x) = x$. These are illustrated in Fig. 1.7.1 and clearly satisfy the required conditions. Since we are dealing with only 1 degree of freedom, the matrix paraphernalia collapses as follows:

$$K = [K_{11}] = K_{11} \quad (1.7.1)$$

$$F = \{F_1\} = F_1 \quad (1.7.2)$$

$$d = \{d_1\} = d_1 \quad (1.7.3)$$

$$K_{11} = a(N_1, N_1) = \int_0^1 \underbrace{N_{1,x}}_{-1} \underbrace{N_{1,x}}_{-1} dx = 1 \quad (1.7.4)$$

$$\begin{aligned} F_1 &= (N_1, f) + N_1(0)h - a(N_1, N_2)q \\ &= \int_0^1 (1-x)f(x) dx + h - \int_0^1 \underbrace{N_{1,x}}_{-1} \underbrace{N_{2,x}}_{+1} dx q \\ &= \int_0^1 (1-x)f(x) dx + h + q \end{aligned} \quad (1.7.5)$$

$$d_1 = K_{11}^{-1} F_1 = F_1 \quad (1.7.6)$$

Consequently

$$u^h(x) = \underbrace{\left[\int_0^1 (1-y)f(y) dy + h + q \right]}_{d_1} (1-x) + qx \quad (1.7.7)$$

In (1.7.7), y plays the role of a dummy variable. An illustration of (1.7.7) appears in Fig. 1.7.2. To get a feel for the nature of the approximation, let us compare (1.7.7) with the exact solution (see (1.2.3)). It is helpful to consider specific forms for f .

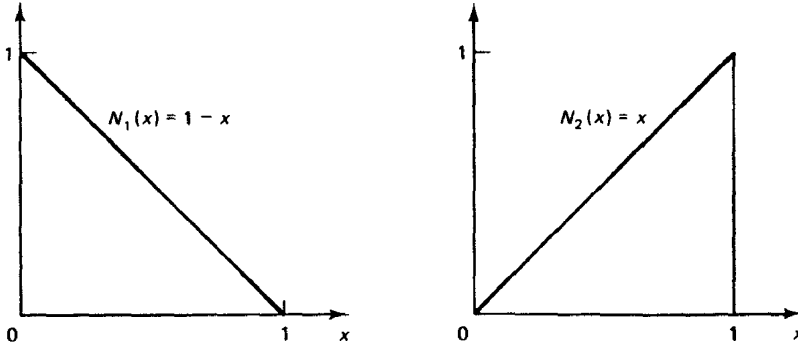


Figure 1.7.1 Functions for the 1 degree of freedom examples. (These functions are secretly the simplest finite element interpolation functions in a one-element context.)

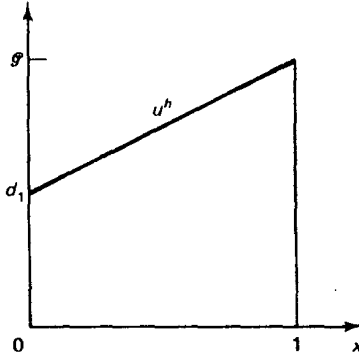


Figure 1.7.2 The Galerkin solution for the 1 degree of freedom example.

i. Let $\ell = 0$. Then

$$u^h(x) = u(x) = q + (1 - x)/\ell \quad (1.7.8)$$

That is, the approximate solution is exact. In fact, it is clear by inspecting (1.7.7) and (1.2.3) that the homogeneous solution (i.e., the part of the solution corresponding to $\ell = 0$) is always exactly represented. The only approximation pertains to the particular solution (i.e., the part of the solution corresponding to $\ell \neq 0$).

ii. Now let us introduce a nonzero ℓ . Assume $\ell(x) = p$, a constant. Then the particular solutions take the form

$$u_{\text{part}}(x) = \frac{p(1 - x^2)}{2} \quad (1.7.9)$$

and

$$u_{\text{part}}^h(x) = \frac{p(1 - x)}{2} \quad (1.7.10)$$

Equations (1.7.9) and (1.7.10) are compared in Fig. 1.7.3. Note that u_{part}^h is exact at $x = 0$ and $x = 1$ and that $u_{\text{part},x}^h$ is exact at $x = \frac{1}{2}$. (It should be clear that it is impossible for u_{part}^h to be exact at all x in the present circumstances. The exact solution, (1.7.9), contains a quadratic term in x , whereas the approximate solution is restricted to linear variation in x by the definitions of N_1 and N_2 .)

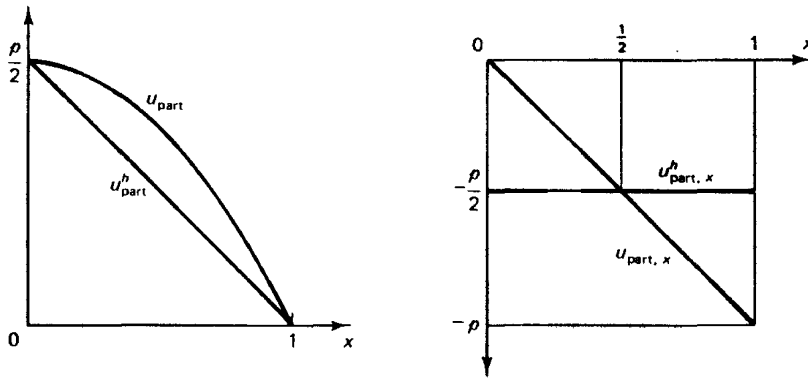


Figure 1.7.3 Comparison of exact and Galerkin particular solutions, Example 1, case (ii).

iii. This time let $\ell(x) = qx$, where q is a constant. This choice for ℓ leads to

$$u_{\text{part}}(x) = \frac{q(1-x^3)}{6} \quad (1.7.11)$$

and

$$u_{\text{part}}^h(x) = \frac{q(1-x)}{6} \quad (1.7.12)$$

which are compared in Fig. 1.7.4. Again we note that the u_{part}^h is exact at $x = 0$ and $x = 1$. There is one point, $x = 1/\sqrt{3}$, at which $u_{\text{part}, x}^h$ is exact.

Let us summarize what we have observed in this example:

- The homogeneous part of u^h is exact in all cases.
- In the presence of nonzero ℓ , u^h is exact at $x = 0$ and $x = 1$.
- For each case, there is at least one point at which $u_{,x}^h$ is exact.

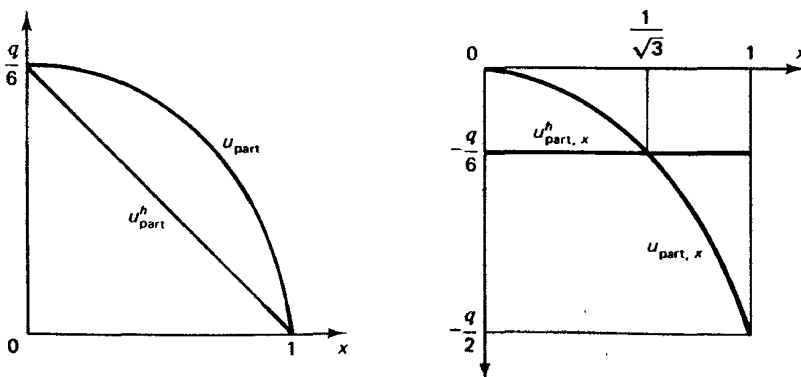


Figure 1.7.4 Comparison of exact and Galerkin particular solutions, Example 1, case (iii).

Example 2 (2 degrees of freedom)

In this case $n = 2$. Thus $w^h = c_1 N_1 + c_2 N_2$, where $N_1(1) = N_2(1) = 0$, and $u^h = d_1 N_1 + d_2 N_2 + q N_3$, where $N_3(1) = 1$. Let us define the N_A 's as follows

$$N_1(x) = \begin{cases} 1 - 2x & 0 \leq x \leq \frac{1}{2} \\ 0 & \frac{1}{2} \leq x \leq 1 \end{cases} \quad (1.7.6)$$

$$N_2(x) = \begin{cases} 2x & 0 \leq x \leq \frac{1}{2} \\ 2(1 - x) & \frac{1}{2} \leq x \leq 1 \end{cases} \quad (1.7.7)$$

$$N_3(x) = \begin{cases} 0 & 0 \leq x \leq \frac{1}{2} \\ 2x - 1 & \frac{1}{2} \leq x \leq 1 \end{cases} \quad (1.7.8)$$

The shape functions are illustrated in Fig. 1.7.5. Typical $w^h \in \mathcal{V}^h$ and $u^h \in \mathcal{S}^h$ and their derivatives are shown in Fig. 1.7.6. Since $n = 2$, the matrix paraphernalia takes the following form:

$$K = \begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix} \quad (1.7.9)$$

$$F = \begin{Bmatrix} F_1 \\ F_2 \end{Bmatrix} \quad (1.7.10)$$

$$d = \begin{Bmatrix} d_1 \\ d_2 \end{Bmatrix} \quad (1.7.11)$$

$$K_{AB} = a(N_A, N_B) = \int_0^1 N_{A,x} N_{B,x} dx = \int_0^{1/2} N_{A,x} N_{B,x} dx + \int_{1/2}^1 N_{A,x} N_{B,x} dx \quad (1.7.12)$$

$$K_{11} = 2, \quad K_{12} = K_{21} = -2, \quad K_{22} = 4 \quad (1.7.13)$$

$$K = 2 \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \quad (1.7.14)$$

$$\begin{aligned} F_A &= (N_A, f) + N_A(0)h - a(N_A, N_3)q \\ &= \int_0^1 N_A f dx + N_A(0)h - \int_{1/2}^1 N_{A,x} N_{3,x} dx q \end{aligned} \quad (1.7.15)$$

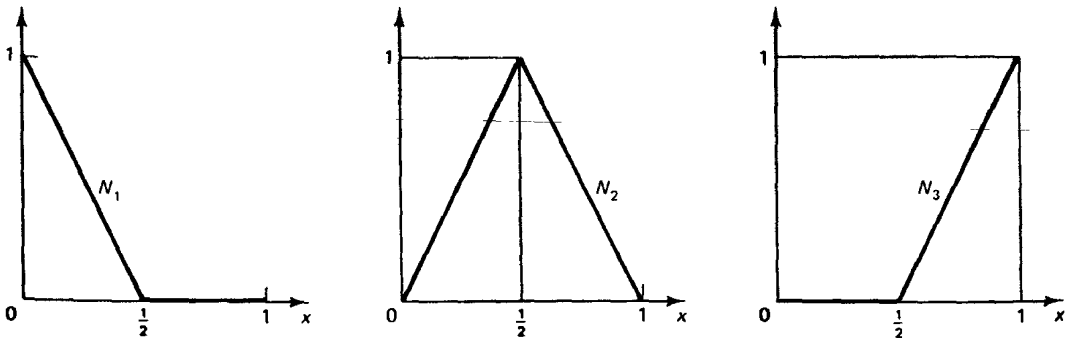


Figure 1.7.5 Functions for the 2 degree of freedom examples. (These functions are secretly the simplest finite element functions in a two-element context.)

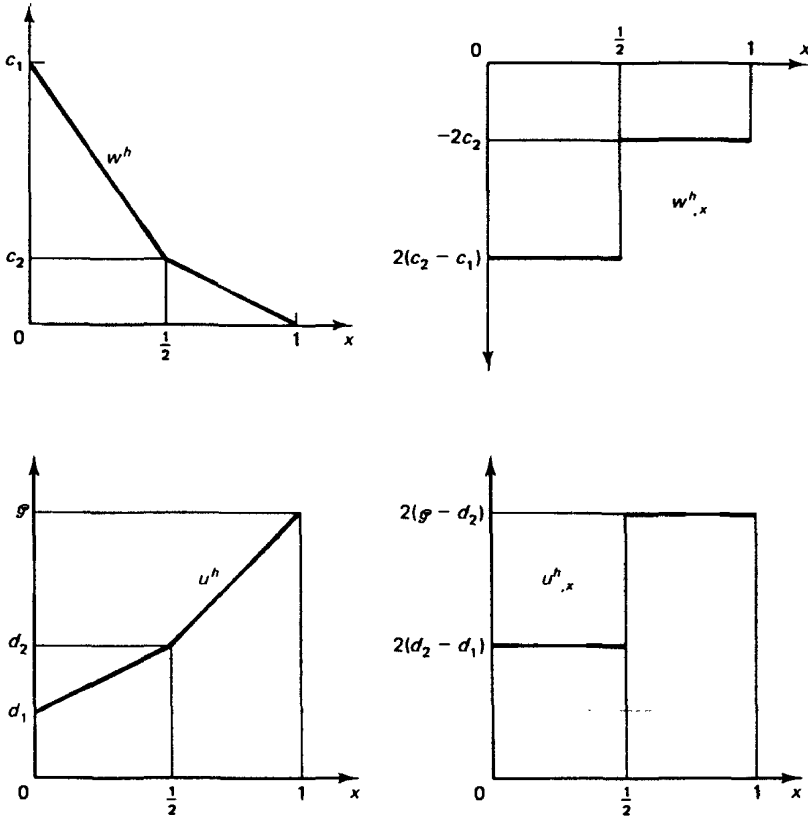


Figure 1.7.6 Typical weighting function and trial solution for the 2 degree of freedom example.

$$F_1 = \int_0^{1/2} (1 - 2x)\ell(x) dx + h \quad (1.7.16)$$

$$F_2 = 2 \int_0^{1/2} x\ell(x) dx + 2 \int_{1/2}^1 (1 - x)\ell(x) dx + 2g \quad (1.7.17)$$

Note that due to the shape functions' discontinuities in slope at $x = \frac{1}{2}$, it is convenient to express integrals over the subintervals $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$ (e.g., see (1.7.12) and (1.7.15)). We need not worry about the value of the derivative of N_A at $x = \frac{1}{2}$ (it suffers a discontinuity there and thus is not well-defined classically) since it has no effect on the integrals in (1.7.12). This amounts to employing the notion of a **generalized derivative**.

We shall again analyze the three cases considered in Example 1.

i. $\ell = 0$.

$$\mathbf{F} = \begin{Bmatrix} h \\ 2g \end{Bmatrix} \quad (1.7.18)$$

$$\mathbf{d} = \mathbf{K}^{-1} \mathbf{F}$$

$$\begin{aligned}
 &= \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{Bmatrix} h \\ 2q \end{Bmatrix} \\
 &= \begin{Bmatrix} q + h \\ q + \frac{h}{2} \end{Bmatrix}
 \end{aligned} \tag{1.7.19}$$

This results in

$$\begin{aligned}
 u^h &= (q + h)N_1 + \left(q + \frac{h}{2}\right)N_2 + qN_3 \\
 &= q(N_1 + N_2 + N_3) + h\left(N_1 + \frac{N_2}{2}\right)
 \end{aligned} \tag{1.7.20}$$

$$u^h(x) = q + h(1 - x) \tag{1.7.21}$$

Again, the exact homogeneous solution is obtained. (The reason for this is that the exact solution is linear, and our trial solution is capable of exactly representing any linear function. Galerkin's method will give the exact answer whenever possible—that is, whenever the collection of trial solutions contains the exact solution among its members.)

ii. $f(x) = p = \text{constant}$.

$$F_1 = \frac{p}{4} + h \tag{1.7.22}$$

$$F_2 = \frac{p}{2} + 2q \tag{1.7.23}$$

$$d = \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{Bmatrix} \frac{p}{4} + h \\ \frac{p}{2} + 2q \end{Bmatrix} = \begin{Bmatrix} \frac{p}{2} + q + h \\ \frac{3p}{8} + q + \frac{h}{2} \end{Bmatrix} \tag{1.7.24}$$

The solution takes the form

$$u^h(x) = q + h(1 - x) + u_{\text{part}}^h(x) \tag{1.7.25}$$

$$u_{\text{part}}^h = \frac{p}{2}N_1 + \frac{3p}{8}N_2 \tag{1.7.26}$$

The approximate particular solution is compared with the exact in Fig. 1.7.7, from which we see that agreement is achieved at $x = 0, \frac{1}{2}$ and 1, and derivatives coincide at $x = \frac{1}{4}$ and $\frac{3}{4}$.

iii. $f(x) = qx, q = \text{constant}$.

$$F_1 = \frac{q}{24} + h \tag{1.7.27}$$

$$F_2 = \frac{q}{4} + 2q \tag{1.7.28}$$

Sec. 1.7 Examples: 1 and 2 Degrees of Freedom

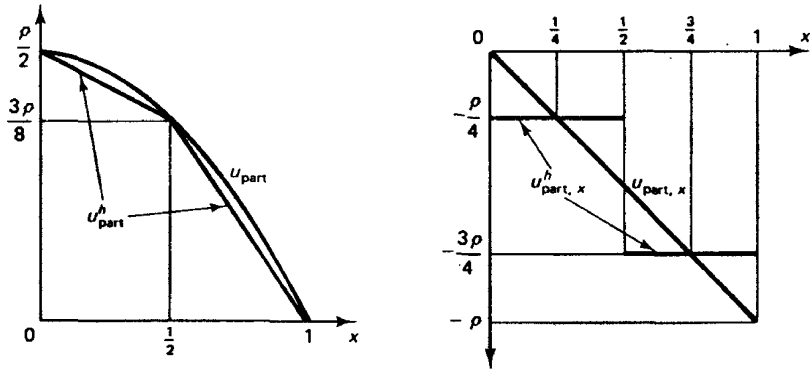


Figure 1.7.7 Comparison of exact and Galerkin particular solutions, Example 2, case (ii).

$$d = \begin{Bmatrix} \frac{q}{6} + q + h \\ \frac{7q}{48} + q + \frac{h}{2} \end{Bmatrix}$$

Again u^h may be expressed in the form (1.7.25), where

$$u_{\text{part}}^h = \frac{q}{6}N_1 + \frac{7q}{48}N_2$$

A comparison is presented in Fig. 1.7.8. The Galerkin solution is seen to be exact again at $x = 0, \frac{1}{2},$ and 1 , and the derivative is exact at two points.

Let us summarize the salient observations of Example 2:

- The homogeneous part of u^h is exact in all cases, as in Example 1. (A proof for this is given after Equation (1.7.21).)

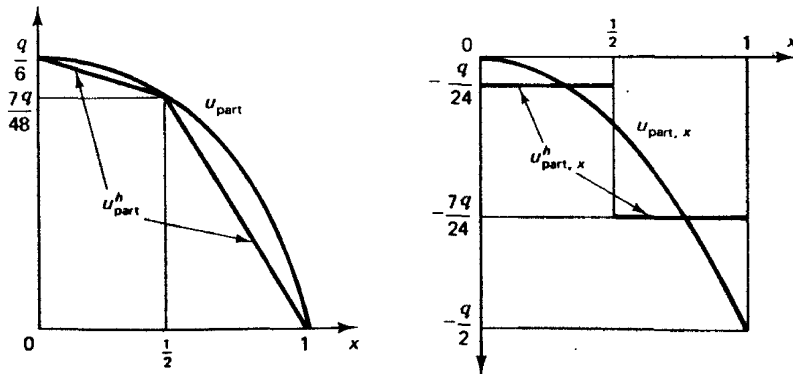


Figure 1.7.8 Comparison of exact and Galerkin particular solutions, Example 2, case (iii).